8.3 MATTER WAVES

8.3.1 Introduction - Wave nature of particles

So far, we learnt that the characteristics of particles and waves are different. A wave is specified by its frequency, wavelength, wave velocity, amplitude and intensity. It spreads out and occupies a relatively large region of space. A particle specified by its mass, velocity, momentum and energy occupies a definite position in space and is very small in size.

Classical physics treated particles and waves as distinct entities. But quantum theory suggested dual character for radiations – that is, radiation behaves as a wave at times and as a particle at other times.

From this wave – particle duality of radiation, the concept of wave nature of matter arises which we will see in this section.

De Broglie wave:

The wave–particle duality of radiation was extended to matter by a French physicist Louis de Broglie (pronounced as de Broy) in 1924.

Greatly influenced by the symmetry in nature, de Broglie suggested that if radiation like light can act as particles at times, then material particles like electrons can also act as waves at times.

According to de Broglie hypothesis, all material particles like electrons, protons, neutrons in motion are associated with waves. These waves are called de Broglie waves or matter waves.

8.3.2 De Broglie wave length:

The momentum of photon of frequency v is given by

$$p = \frac{hv}{c} = \frac{h}{\lambda} \quad \text{since } c = v\lambda$$

The wavelength of a photon in terms of its momentum is

$$\lambda = \frac{h}{p} \tag{8.9}$$

According to de Broglie, the above equation is completely a general one and this is applicable to material particles as well. Therefore, for a particle of mass m travelling with speed υ, the wavelength is given by

$$\lambda = \frac{h}{m\upsilon} = \frac{h}{p} \tag{8.10}$$

This wavelength of the matter waves is known as **de Broglie wavelength**. This equation relates the wave character (the wave length λ) and the particle character (the momentum p) through Planck's constant.

8.3.3 De Broglie wave length of electrons:

Let an electron of mass m be accelerated through a potential difference of V volt. The kinetic energy acquired by the electron is given by

$$\frac{1}{2}m\upsilon^2 = eV$$

Therefore, the speed υ of the electron is

$$\upsilon = \sqrt{\frac{2eV}{m}} \tag{8.11}$$

Hence, the de Broglie wavelength of the matter waves associated with electron is

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